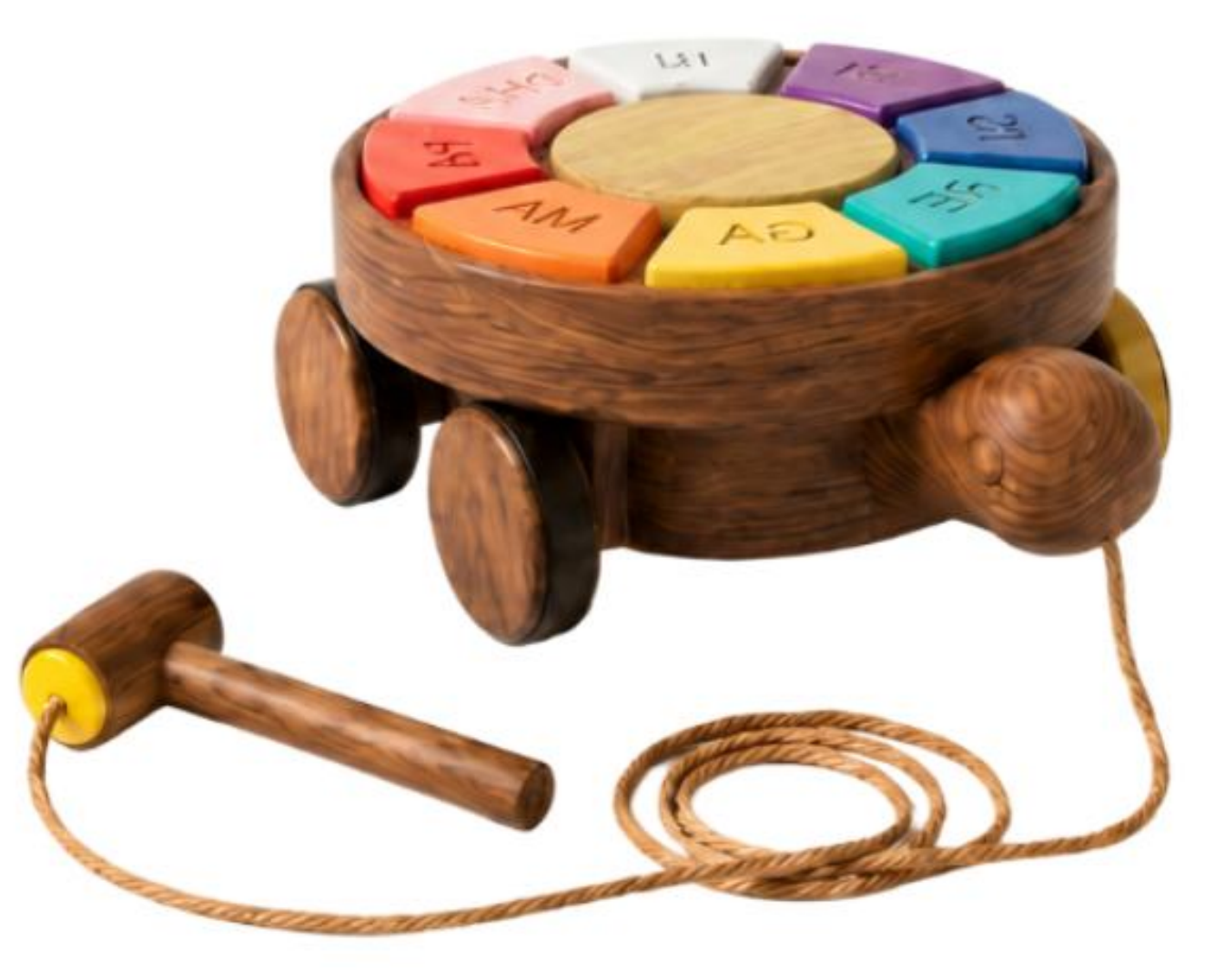


SWARA

PULL TOY

A sustainable musical toy system that gives children physical ownership of individual notes.

Place
Strike
Drum

Context

A child who chooses the note, places it, and strikes it will remember that note — because they made it.The toy combines musical learning, physical interaction and Indian cultural references into a single play experience.

Swara is built around one simple idea:

Reasearch

- Conducted primary research through interviews with 10 parents and a Google survey with 55 respondents.
- Observed children interacting with existing musical toys to understand their play behaviour and interaction patterns.
- Conducted competitive benchmarking of existing musical toys to analyse their features, interactions and learning approaches.
- Did User research by created User Journey Mapping and Empathy Mapping to identify user needs, behaviours and pain points.

01 **Passive Musical Learning**

Make musical learning physically interactive

02 **Limited Sargam Representation**

Introduce the Indian Sa–Re–Ga–Ma–Pa–Da–Ni–Sa scale

03 **Lack of Tactile Learning**

Cost-effective design – One product should serve multiple purposes.

04 **Limited Cause-and-Effect Interaction**

Create an immediate action → sound response

05 **Limited Multi-Modal Play**

Make learning simple through play

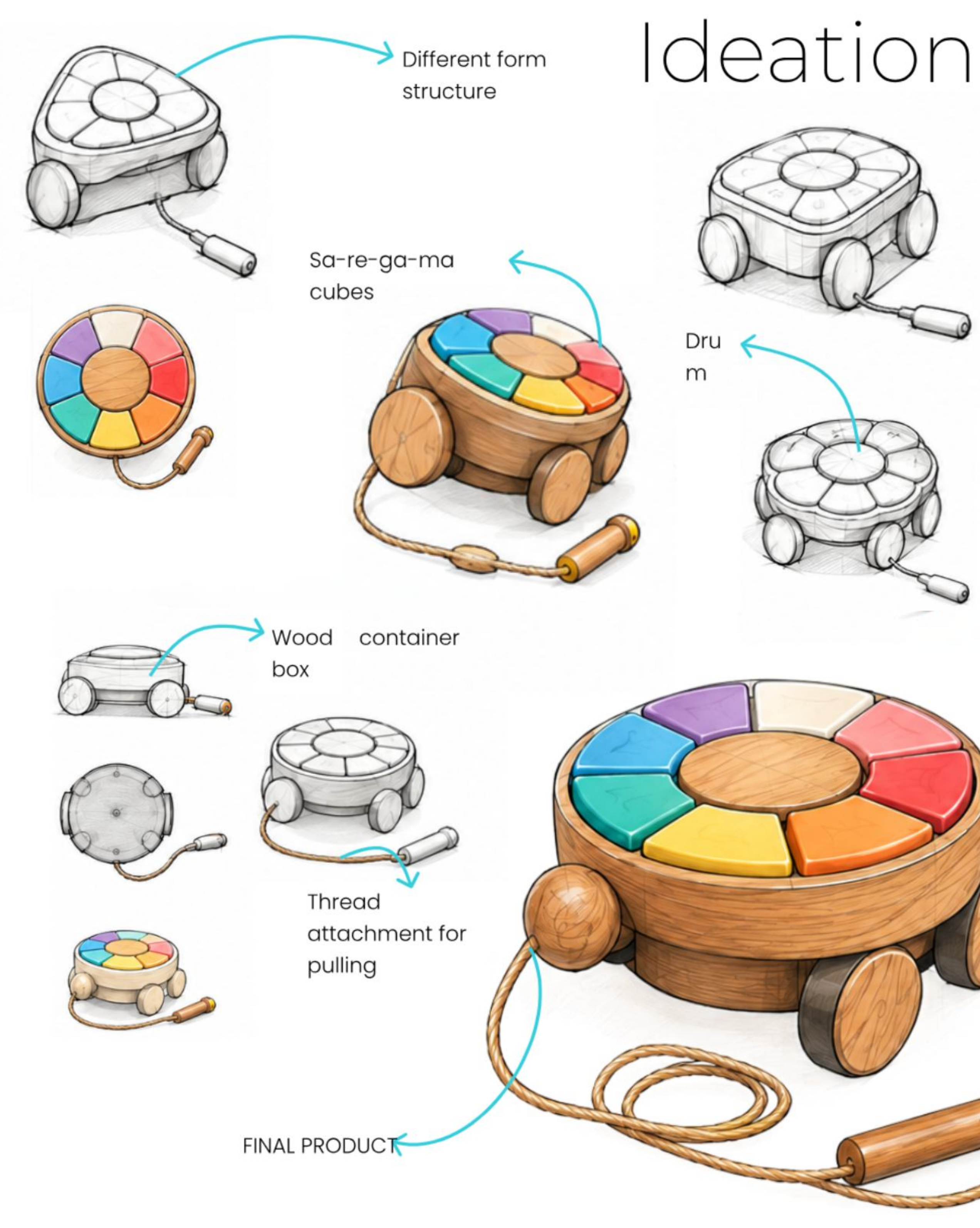
06 **Complex Musical Concepts**

Combine melody · rhythm · movement

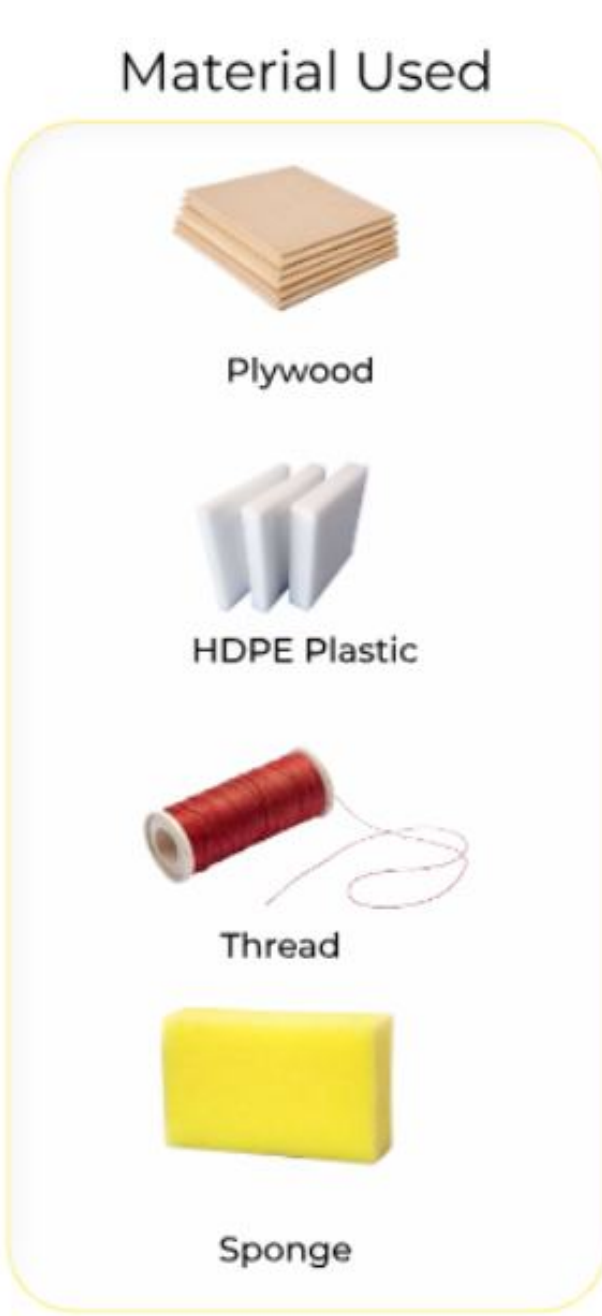
Mood Board

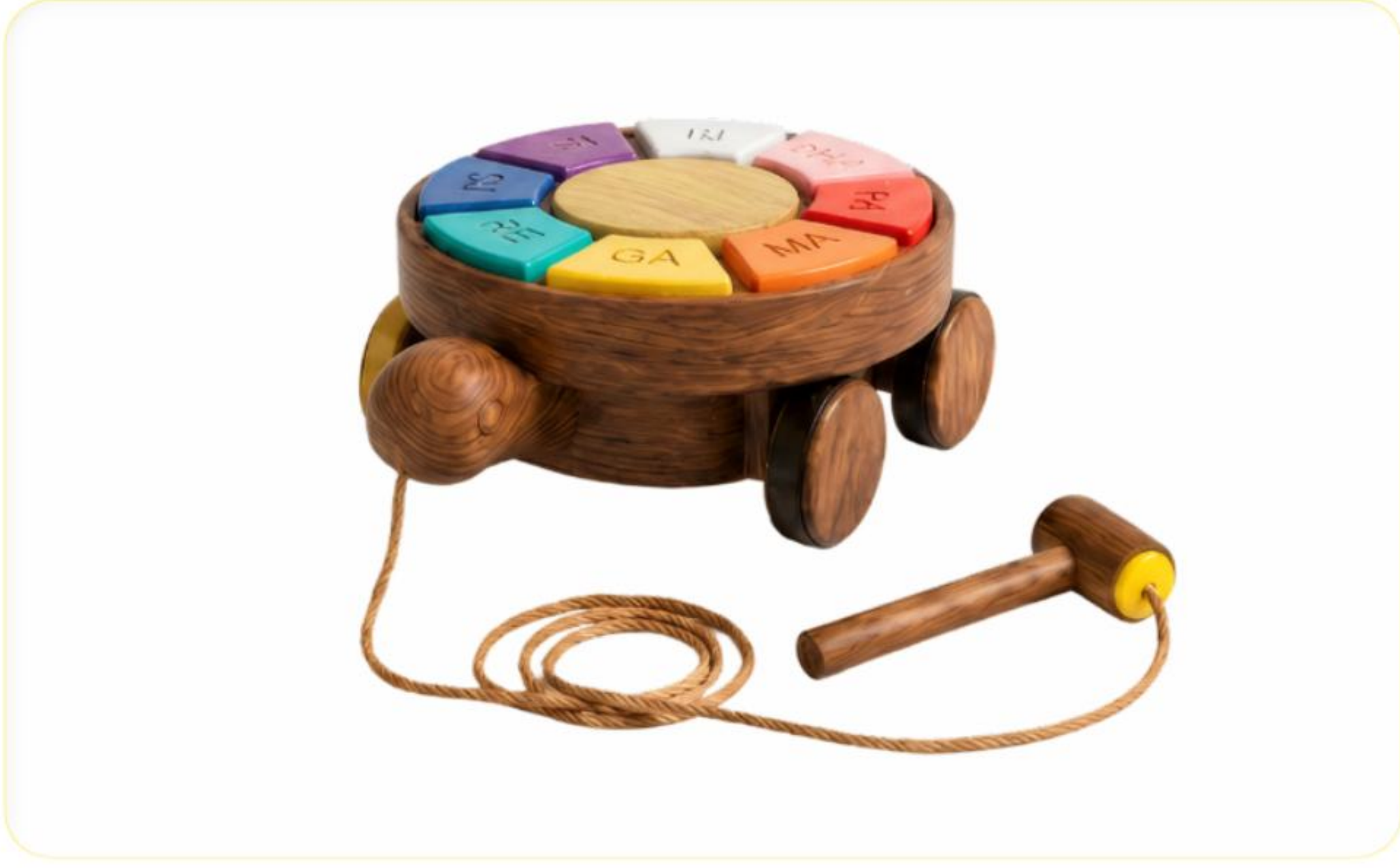
Inspiration
Music
Comfort
Colours
Form

Ideation


Different form structure
Sa-re-ga-ma cubes
Dru m
Wood container box
Thread attachment for pulling
FINAL PRODUCT

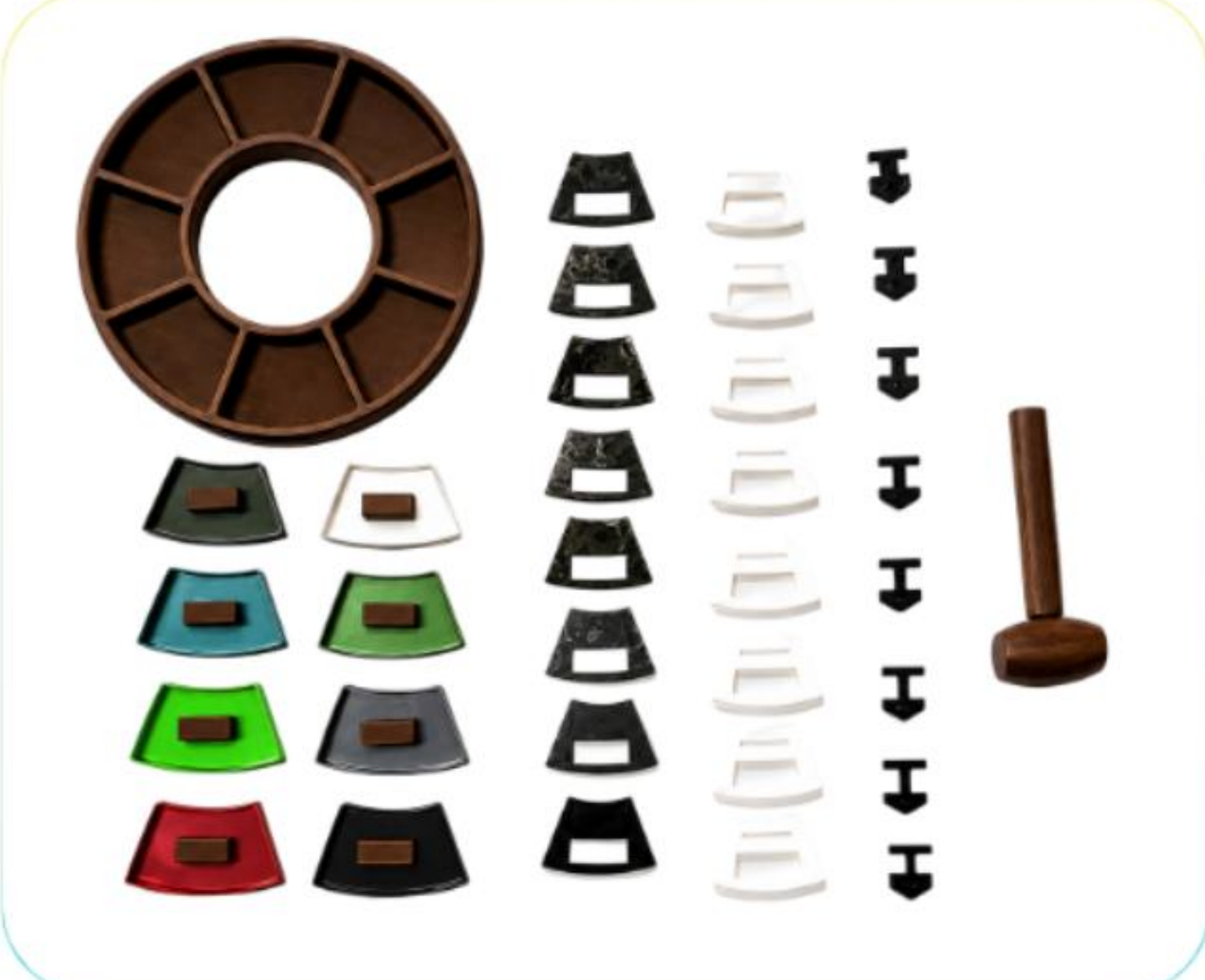
Renders


Material Used
Plywood
HDPE Plastic
Thread
Sponge



Place
Strike
Drum

Prototype



Place

Block fits into the slot and aligns with the metal plate.

Strike

Mallet hits the block, compressing the spring.

Contact

Metal parts collide and create vibration.

Vibrate

Vibration produces a warm, sustained note.

Return

Spring pushes the block back up.

Tuning

Metal size controls pitch: smaller = higher, larger = lower. 8 notes: Sa–Re–Ga–Ma–Pa–Da–Ni–Sa.